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Incidence and Predictors of Adverse Drug Reactions in
Hospitalized Elderly Patients: A Prospective Cohort Study

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Abstract

Objective: To determine the incidence and predictors of adverse drug reactions (ADRs) in hospitalized elderly patients.

Methods: Prospective cohort study of 1,247 patients aged 65+ admitted to two teaching hospitals over 12 months.

Results: ADR incidence was 18.3% (228/1,247). Most common agents: anticoagulants (23%), antibiotics (19%), opioids (15%).

Independent predictors: polypharmacy (OR 2.8), renal impairment (OR 2.1), history of ADR (OR 3.4), age >80 (OR 1.7).

Case Vignette - Fictional Patient

Patient: George P. Harrison, 82-year-old male

Admission: Hip fracture repair

Medications on Admission:

1. Warfarin 3mg daily (for atrial fibrillation)
2. Digoxin 0.125mg daily

3. Furosemide 40mg daily
4. Amlodipine 5mg daily
5. Omeprazole 20mg daily
6. Paracetamol 1g QDS PRN

ADR Event: On day 4 post-surgery, patient developed confusion, nausea, and visual disturbances. Digoxin level: 2.8 ng/mL (therapeutic: 0.5-2.0).

Root Cause: Acute kidney injury from post-surgical dehydration reduced digoxin clearance. Furosemide dose not adjusted.

Management: Digoxin withheld, IV fluids initiated, renal function monitored. Recovery in 48 hours.

Lesson: Drug level monitoring essential when renal function changes.

Conclusions

ADRs are common in elderly inpatients and significantly prolong hospital stay. Polypharmacy and renal impairment are major predictors. Pharmacist-led medication review reduces ADR risk.